DSC 640
Mahad Farah
Analysis Summary & Communication Plan: National Database of Childcare Prices

Summary of Analysis

The National Database of Childcare Prices (NDCP) provides detailed estimates of childcare costs across U.S. counties between 2008 and 2018. The data is broken down by state, county, care type (center-based vs. family-based), age group (infant, toddler, preschool), and percentile pricing (median and 75th percentile). Each cost variable is supplemented with a quality flag indicating the reliability of the estimate.

Using this dataset, I explored trends in pricing growth over time, regional disparities in childcare aSordability, and the higher burden of infant care costs. Several patterns emerged that can help inform policymakers, families, and researchers.

Key Findings

Infant care is the most expensive: Center-based infant care in high-cost regions regularly exceeds $1,500/month. Costs increase as children get younger, making care least aSordable when families have the fewest financial resources.

Large geographic disparities: Cost diSerences between states can exceed $1,000/month. Urban counties such as those in California, Massachusetts, New York consistently rank highest.

Costs rising faster than wages: Between 2008–2018, infant care costs increased every year nationally. Salary growth did not keep pace, especially for single-parent households.

Family-based care oSers some relief: Even the “cheaper” option can consume a substantial share of household income. These patterns aligned with expectations from Milestone 2’s medium planning Milestone 2 and informed the visuals developed in Milestone 3.

Assumptions

Monthly childcare costs may not be inflation adjusted. Potentially underestimating real change.

Columns ending in _flag indicate data reliability tiers. I assumed 1 = highest confidence, 3 = modeled values.

Center-based (MCInfant) and family-based (MFCCInfant) infant cost columns meaningfully reflect market prices.

Missing values treated as unavailable, not zero, preventing distorted cost averages.

Items Needing Clarification

Exact flag definitions from the NDCP would improve filtering confidence.
Urban vs rural designation is implied but not explicitly provided.
Licensing diSerences across states could influence cost comparability.
Lack of inflation adjustment documentation limits economic interpretation across years.

Direction of Story / Message

This narrative seeks to highlight the increasing and disproportionate financial strain of childcare expenses throughout the U.S. The communication will emphasize:

The rising expenses of childcare for infants and their unequal impact on families with low incomes.

Comparisons between states and counties that uncover concealed aSordability issues in specific areas.

The role of data transparency and accurate estimates in promoting fair funding and policy assistance.

Visuals and narratives will highlight disparities in access, inflation trends over time, and the financial strain these places on working parents, especially mothers.

Target Audience

Policymakers & Legislators: Budget allocation decisions & subsidy programs.

Advocacy Organizations & Parents: Evidence to support reform.
Researchers & Journalists: Insights on economic and demographic disparities. Each medium was tailored to meet these audiences information access needs.

Mediums Included & Why

(From Milestones 2 & 3)

Interactive Dashboard (Tableau/Power BI): Best for policymakers and journalists who need filtering and exploration.

PowerPoint Presentation: Narrative guidance for formal briefings, academic review, and stakeholder communication.

These mediums connect analytical rigor with accessibility and persuasion.

Design Decisions

Gestalt principles (alignment, proximity, clear hierarchy) guided layouts.

Color gradients are used on maps to highlight regional inequality.

Icons and labels clarify FLAG reliability levels to prevent misleading conclusions.

Annotation callouts ensure the audience's eyes are guided to key takeaways.

The focus was on clarity, not decoration. Leading with questions that the audience needs answered.

Ethical Considerations

Data Changes Made
Cleaned $ and commas from cost columns
Removed rows missing all infant cost values
Converted numeric columns for trend visualization
Created derived fields (Average infant cost, aSordability ratio)

Lessons Learned

Data wrangling must be well documented to prevent misinterpretation. Designing for audience accessibility is just as important as producing analysis. Data flags and metadata are crucial. Ignoring them degrades trust.

What I Would Do DiGerently

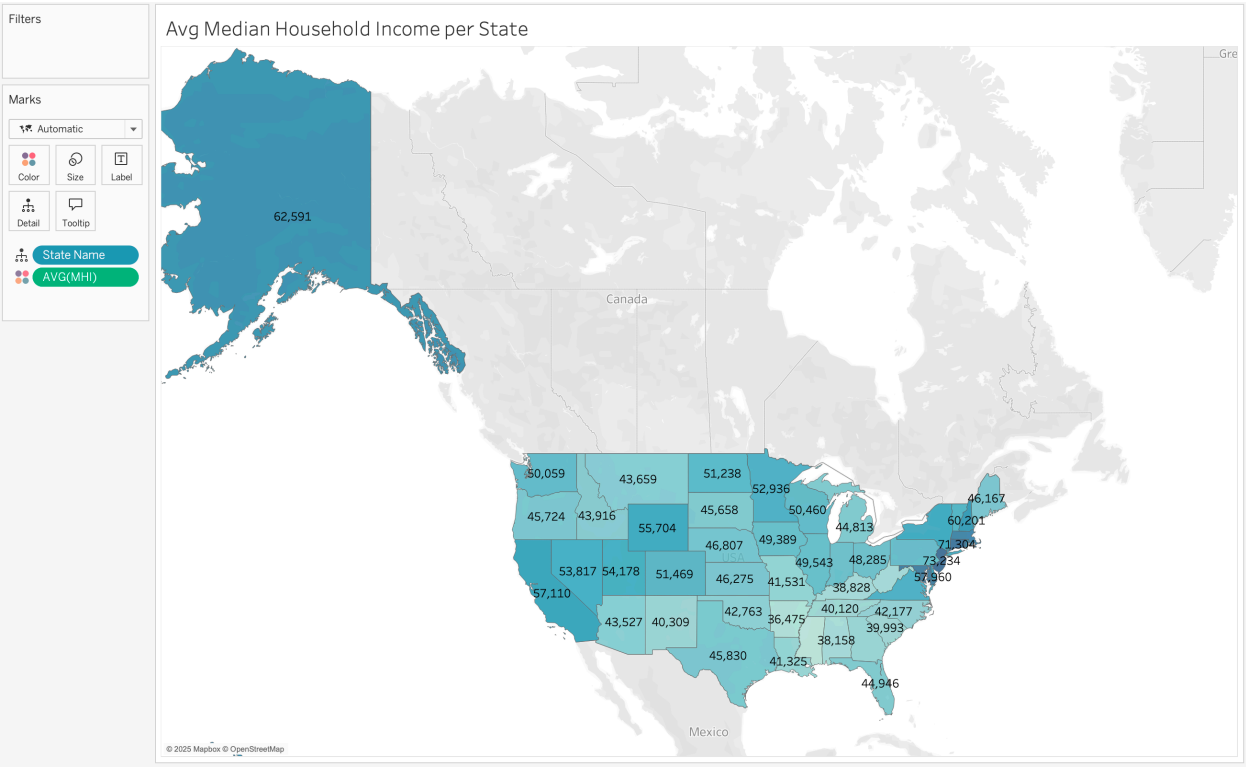
Spend more time exploring county level inequalities using aSordability mapping.
Investigate inflation adjustments to strengthen economic claims.
Incorporate qualitative voices of parents to deepen the advocacy impact.

What I Enjoyed Most

I most enjoyed transforming a complex federal dataset into an accessible narrative that has real-world policy relevance. This project illustrated how analytics can inform critical social issues and how powerful storytelling can drive change. It cemented my confidence in communicating data for impact.

Mediums

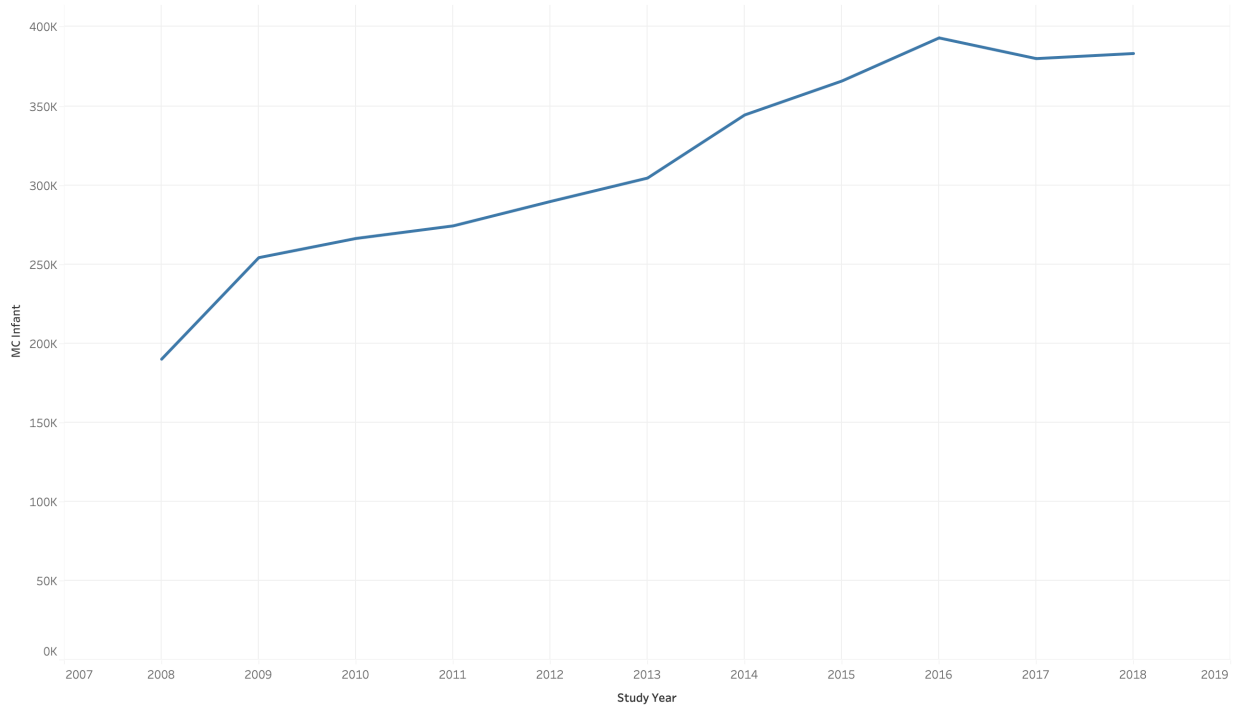
Medium 1: Tableau



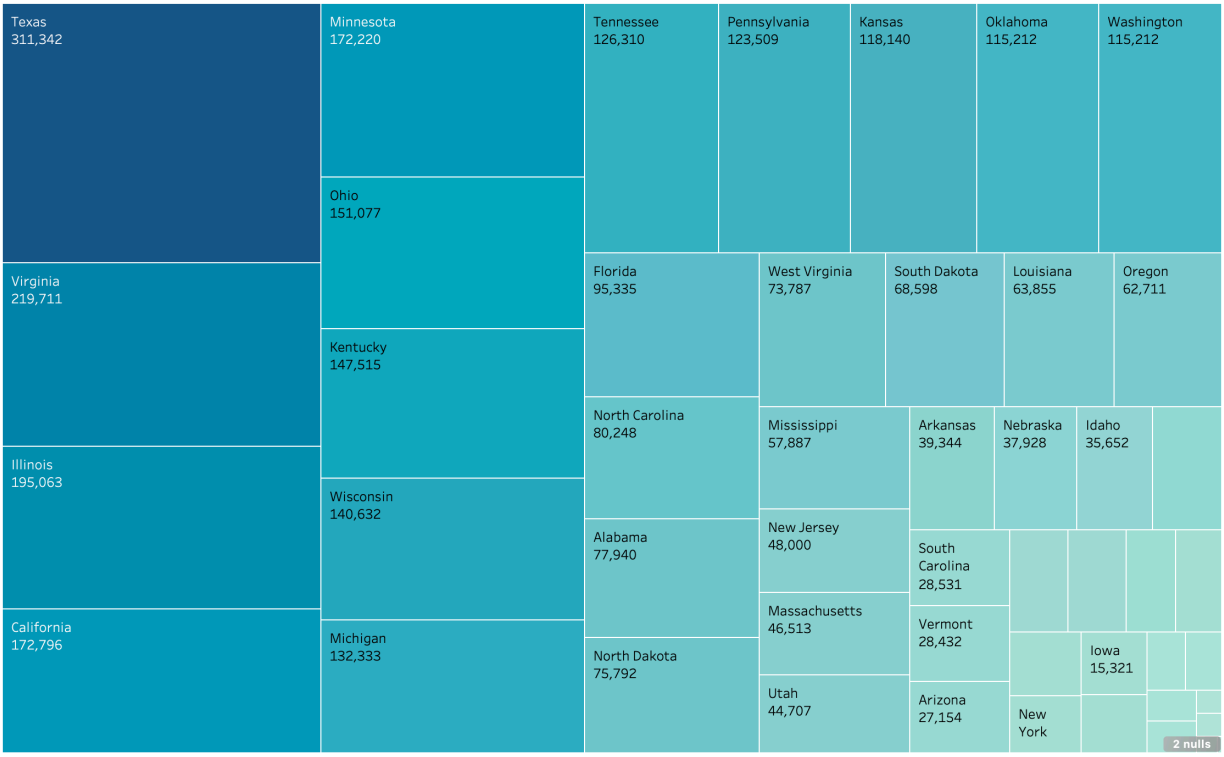
Columns: Study Year

Rows: SUM(MC Infant)

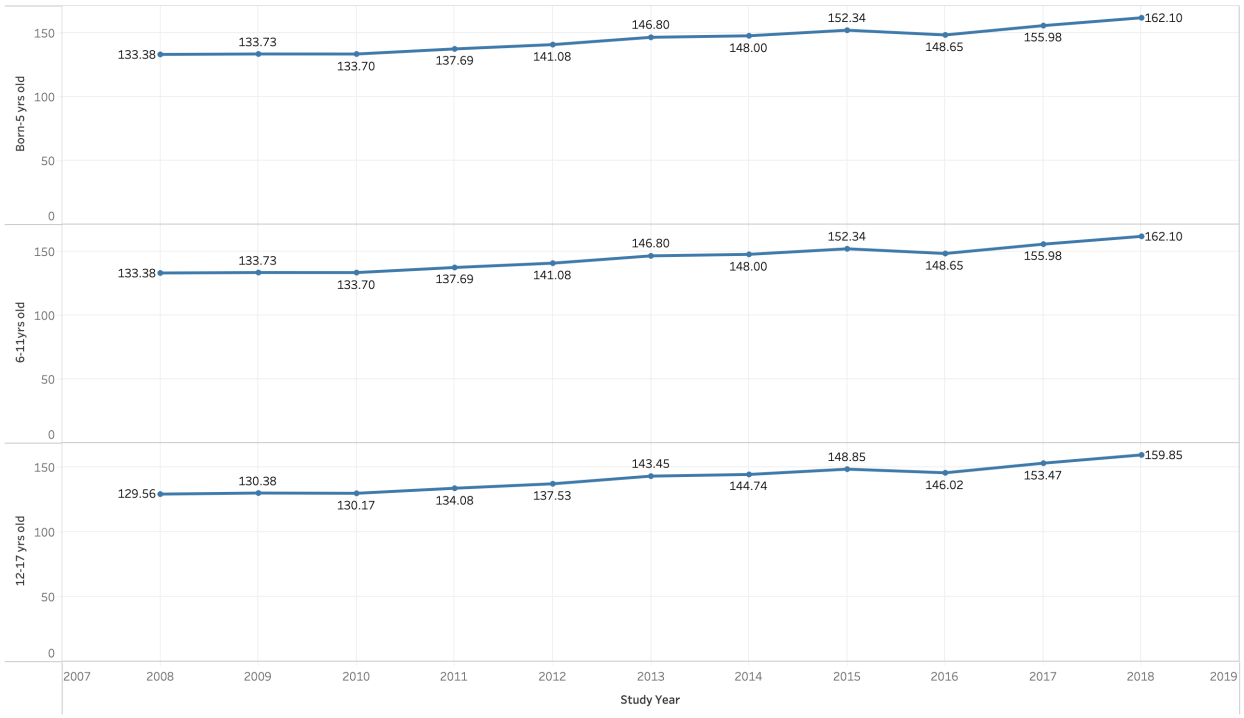
Median Cost for Infants per Year



Median Cost for Infants per Year



Cost comparison by age group over a decade.



Medium 2: Infographic

This infographic was created for an audience of advocates, parents, and grant writers. People who need a portable, informative visual to support their childcare reform efforts. The design prioritizes clarity and emotional impact by focusing on four high-value insights: the extreme cost of infant care in some counties, geographic disparities shown through a U.S. map, some differences between care types, and long-term price trends. Each section is visually distinct yet cohesive, guiding the viewer from headline statistics to deeper comparisons. What you should get out of this is answering your main questions like “what’s happening” (key facts), “where it’s happening” (map), “what drives it” (care type), and “how it’s changing” (trends).

Gestalt principles informed key layout decisions. Proximity and alignment group related visuals and text for easy scanning. Sizing and hierarchy help emphasize the most impactful numbers like infant care exceeding a higher amount depending on the state, using bold, oversized boxes at the top left for the highest costs. These callouts help summarize the urgency of differing costs and offers a clear takeaway for decision-makers. Overall, this

distills complex data into visual language that is approachable, resonanting, and ready for distribution across advocacy channels or policy briefings.

Medium 3: Powerpoint Slide

Background

Childcare is one of the largest household expenses for families, often exceeding rent or tuition. When care is unaffordable, parents, especially mothers, may leave the workforce, deepening gender and income gaps. Affordable care is crucial for child development, economic mobility, and labor market participation.

The National Database of Childcare Prices (NDCP) includes cost estimates from over 3,000 U.S. counties between 2008 and 2018. It breaks down prices by care type (center-based vs. family-based), child age (infant, toddler, preschool), and percentile levels (median and 75th). Flags indicate data quality and estimation methods.

Key Findings

Trends Over Time

Line graph (Infant care, 2008–2018):
A time-series chart shows a steady upward trend in infant care costs across all states.

Key takeaway:
Infant care costs have risen faster than the rate of inflation in many regions, particularly in family-based care settings. This widening gap disproportionately affects lower-income and single-parent households.

Implications and Recommendations

Universal affordability: Infant care is financially unsustainable in many areas. Direct subsidies or tax credits are essential.

Support rural providers: Many areas with low costs also suffer from low availability; investments are needed to expand access.

Improve data transparency: Requiring standardized reporting across states can enhance monitoring and drive smarter funding.

Work Cited

U.S. Department of Labor. (2022). *National Database of Childcare Prices*

U.S. Census Bureau. (2008–2018). Median household income estimates

Bureau of Labor Statistics. (CPI, Inflation Index)